

Supplementary Information: Long-term citizen science data reveals environmental correlates of tropical tree flowering at the regional scale (Anujan et al., *TBD*)

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Section S1. Species descriptions and SeasonWatch observations

Table S1. Species descriptions.**Taxonomic description, appearance of tree and reproductive parts, phenology in Kerala and description of the SeasonWatch dataset for jackfruit, mango and tamarind

	Jackfruit	Mango	Tamarind
Accepted Scientific name	*Artocarpus heterophyllus*	*Mangifera indica*	T*amarindus indica*
Name in Malayalam	Pilavu/Plavu (tree), chakka (fruit)	Maavu/Moochi (tree), manga (fruit)	Puli, Valan puli, Kolpuli
Family	Moraceae	Anacardiaceae	Fabaceae
Description (Trees of Delhi)	A handsome evergreen tree with glossy leaves up to 20 m tall in ideal conditions. Cultivated for its misshapen fruits - largest edible fruit in the world	Semi-evergreen trees with long, glossy leaves that can grow up to 35 m in favourable sites. Bark is grey-brown and rough with shallow cracks.	Large, handsome, long-lived tree with a squat trunk and a shady crown. Introduced to India from E Africa, but virtually indigenous. The bark is dark grey, rough with shallow fissures
Flower description	Male and female separate, on the same tree. Male flowers tiny, in clusters emerging from new branchlets; female flowers densely crowded on old branches or directly from the trunk	Mango flowers are tiny, yellowish-green, strongly scented, in huge, branched clusters. Most flowers are male and the rest bisexual.	Small, with 3 unequal-sized yellow petals veined with red; flower cup has 4 creamy segments
Fruit description	Gigantic, lumpy, barrel- or pear-shaped; pollinated by small flies and beetles	Fruit is smooth-skinned, waxy and in various colours depending on the variety. Flesh is fibrous or pulpy and the stone flattened, kidney-shaped	Beanlike pod bulging over the seeds, up to 20 cm long; green at first, ripening cinnamon-brown. Young pod is covered with a downy felt.
Common varieties in Kerala	Varikka, koozha, etc	Muvandan, chandrakkaran, etc	
Uses	Timber, fruit	Fruit	Fruit, timber
Fruiting and flowering months in Kerala	November to April	January to May	September to April

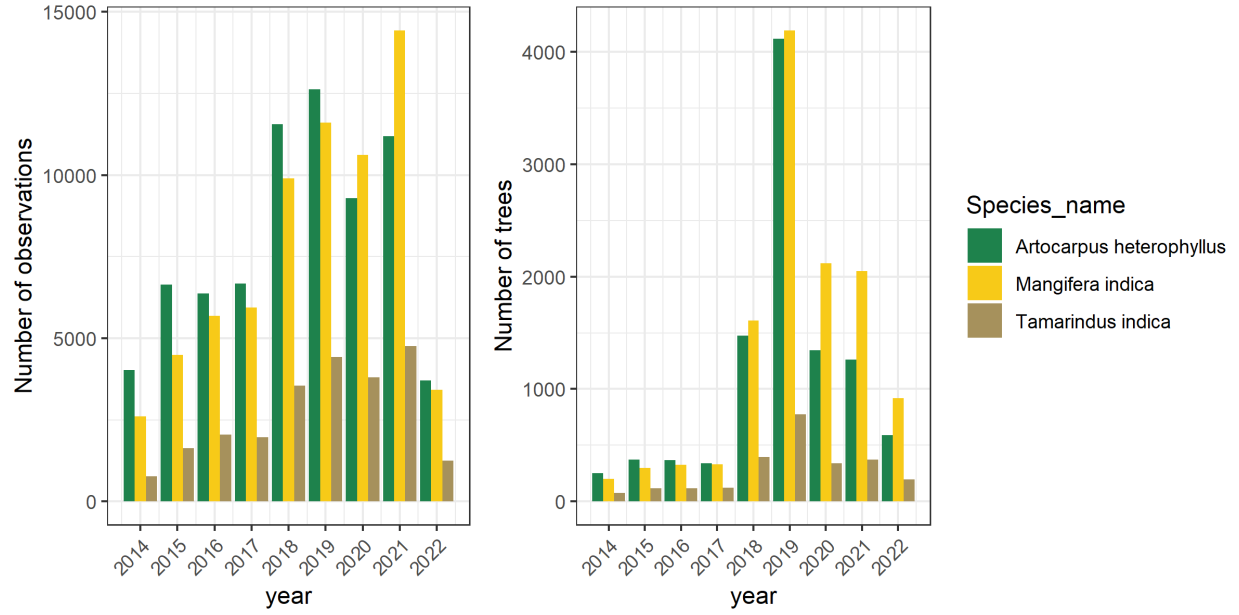


Figure S1. Data distribution of all SeasonWatch observations for the top three species in Kerala across the years. a) Total number of observations including regular and casual b) Number of trees in these observations

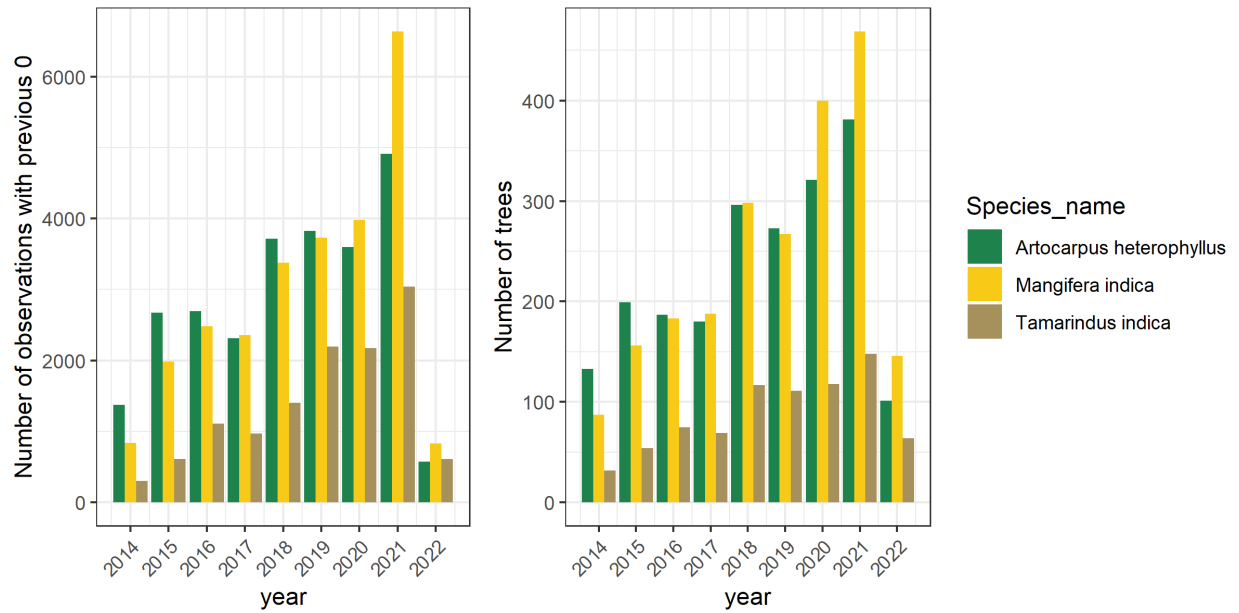


Figure S2. Data distribution of SeasonWatch observations with a 0 flowering observation in the previous week for the top three species in Kerala across the years. a) Total number of observations (includes only regular because casual observations do not have previous week's observation) b) Number of trees in these observations

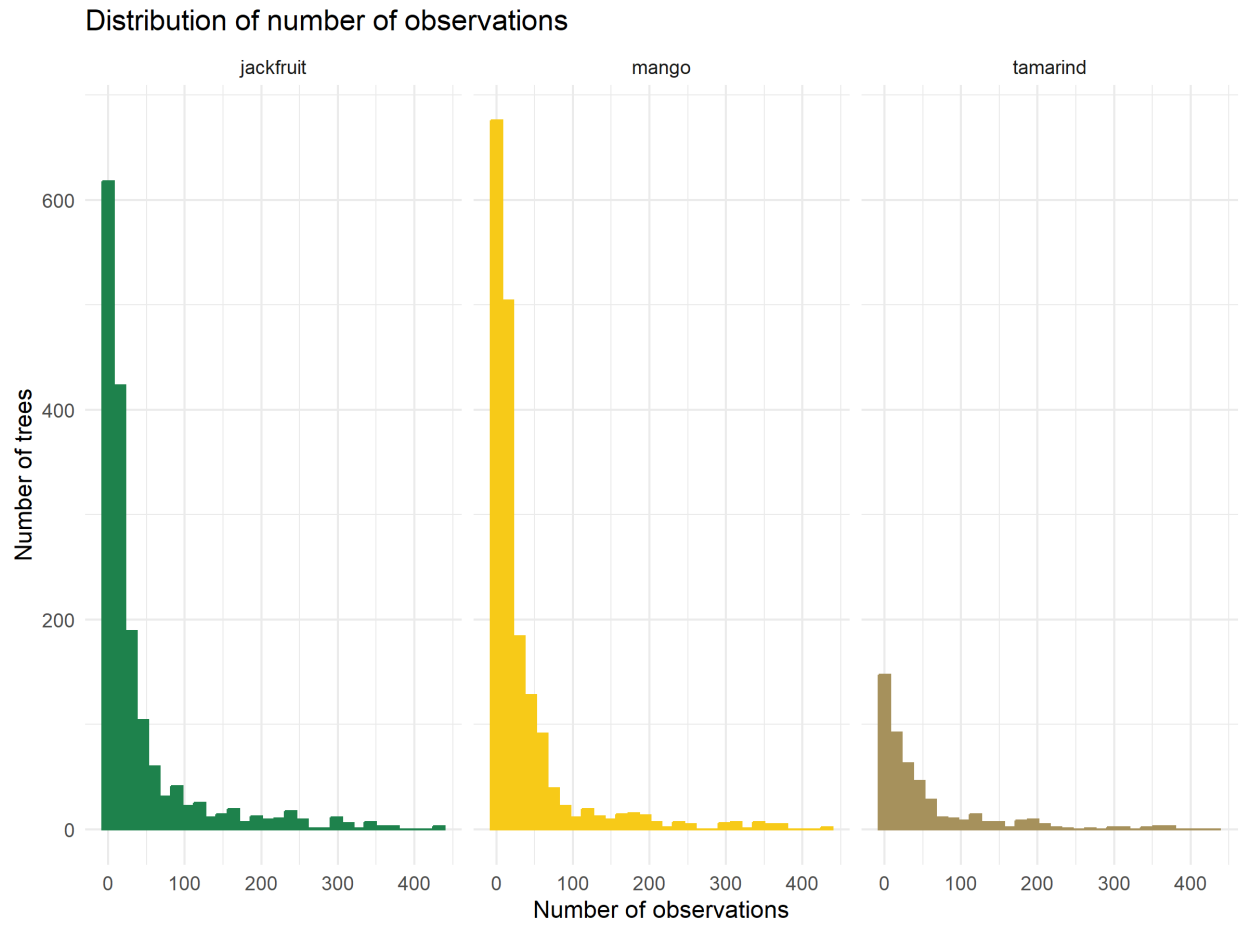


Figure S3. Distribution of regular observations of the top three species in Kerala across the years.

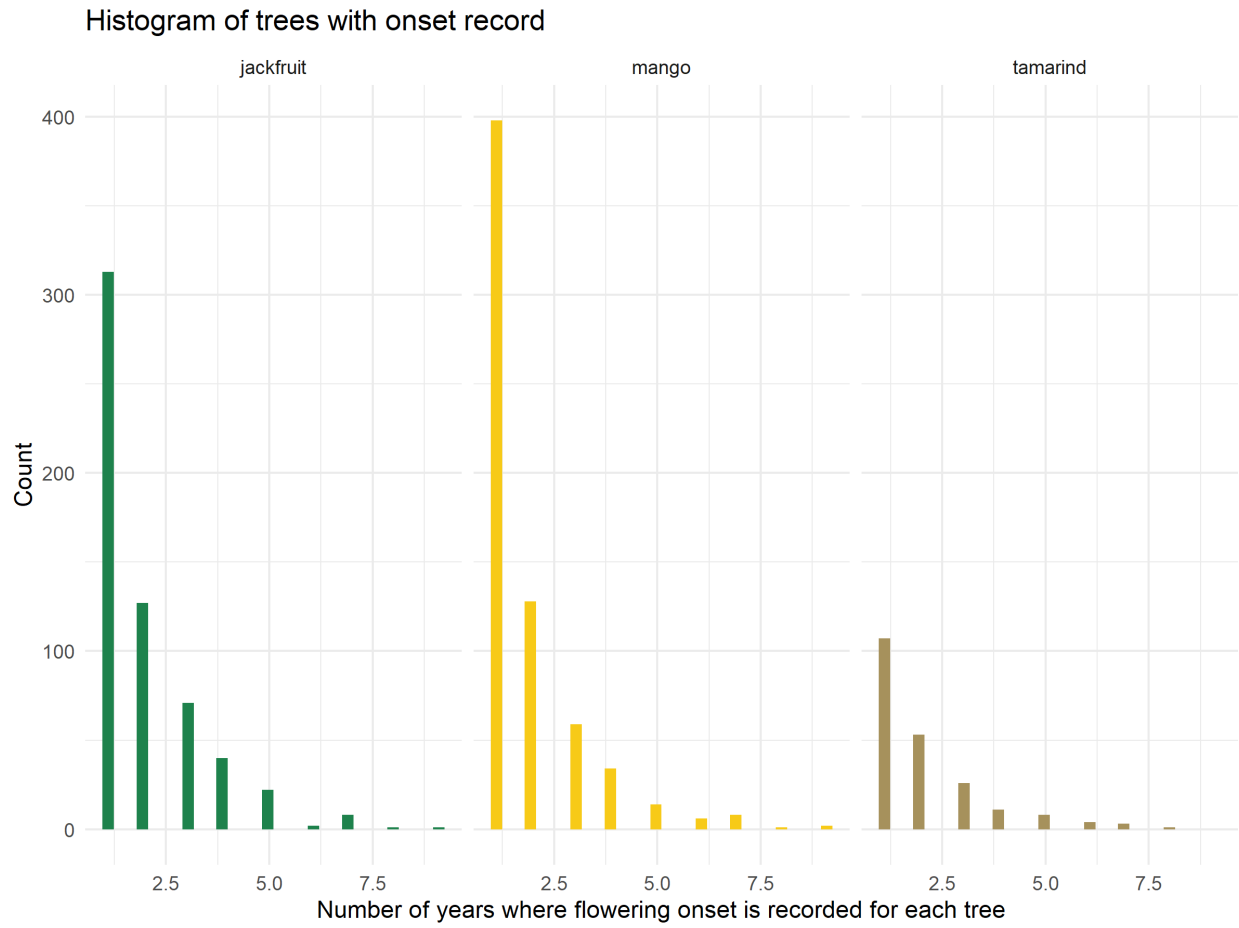


Figure S4. Number of years with observed onset record for each tree

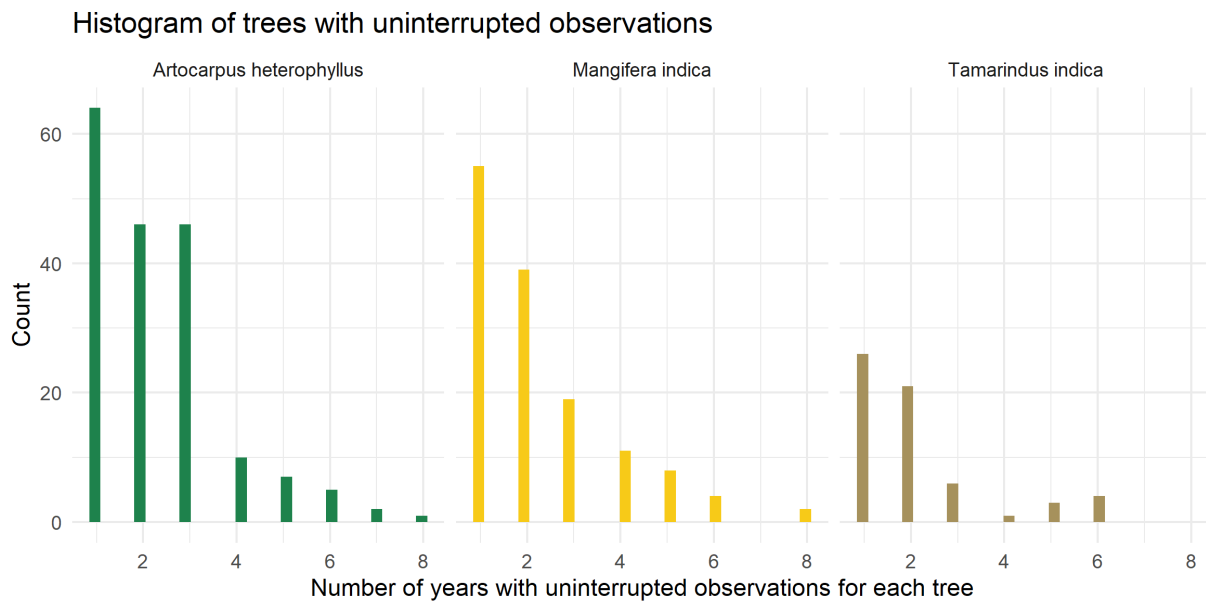
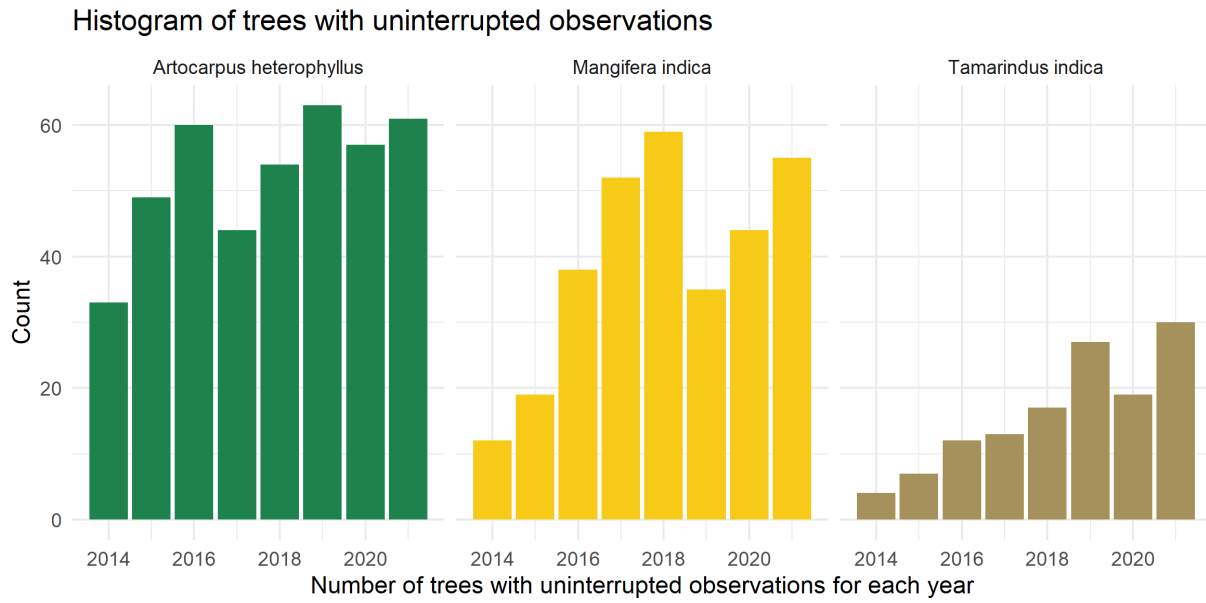


Figure S5. Distribution of trees with uninterrupted flowering record

Section S2. Supplementary methods for data cleaning

Geographic location of points

To validate the geographic location of the data points, we included all the points that were already in Kerala as per the geographic location. From the remaining points, for any that had no location information, we excluded those that were single observations of trees, because of no alternate way to fill these. For regular observations with unconfirmed locations, we filled them with lat/longs from the same tree if available in later observations, or the average lat/long from other trees reported by the same user. As majority of the users were school students observing trees on foot, all regular trees observed by the same user tend to be highly clustered (< 1 km), allowing us to reliably fill an average lat/long from other trees when no observation of a tree had a lat/long. Finally, if no trees by the user had accurate lat/long values and if they were by users who can be located through Google maps (schools or other institutions), we used a geocoding software on Google Sheets, Awesome Table Geocode, to fill these values. We then manually validated these with help from the field coordinator for Kerala. We excluded observations that could not be filled or verified by these methods.

Section S3. Environmental variable methods

Climate Data Column Descriptions

All climate data was calculated on the daily time step for 0.1° pixels and aggregated to the fortnightly, monthly, and quarterly timesteps. A summary of the daily climate data is shown below:

- * PCP = Total daily precipitation (in meters)
- * MinTemp = Minimum daily temperature (in Kelvin)
- * MeanTemp = Average daily temperature (in Kelvin)
- * MaxTemp = Maximum daily temperature (in Kelvin)
- * MeanSM = Average daily soil moisture (the top 1m of the soil column in volumetric soil water content: m³/m³)
- * SolarRad = Total surface net solar radiation (in J/m²)
- * DryDays = Maximum consecutive days with PCP < 1mm (trace amounts are a common artefact in climate reanalysis data)

For more information about these variables (except dry days, which was calculated from PCP), see https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_HOURLY

Each variable has a fortnightly, monthly, and quarterly aggregation. Precipitation and Solar Radiation are accumulated totals (i.e., sum) while the temperature and soil moisture variables are averages. Columns abbreviated “FN” indicate fortnightly, “MN” indicates monthly and “QR” indicates quarterly. For example, “FNMean_MaxTemp” indicates that the mean Maximum Daily Temperature value was taken for the previous fortnight. “QrSum_PCP” indicates that the sum of daily total precipitation was taken for the previous quarter.

Static Data Column Descriptions * light_vals = Static variable. NASA satellite-derived lights dataset as a proxy for urban density

- * elev = Static variable. Elevation above mean sea level derived from 30m SRTM
- * slope = Static variable. Rise/run of elevation dataset
- * aspect = Static variable. Direction of slope

Observation-Level Data

The dataset includes the Grid_ID and Fortnight they belong to, the observation ID, date of observation, species name, latitude, longitude, the response variables (fl_binary, fl_intensity) and the climate indicators. FN: Previous 14 days before date of observation MN: Previous 30 days before date of observation Qr: Previous 90 days before date of observation

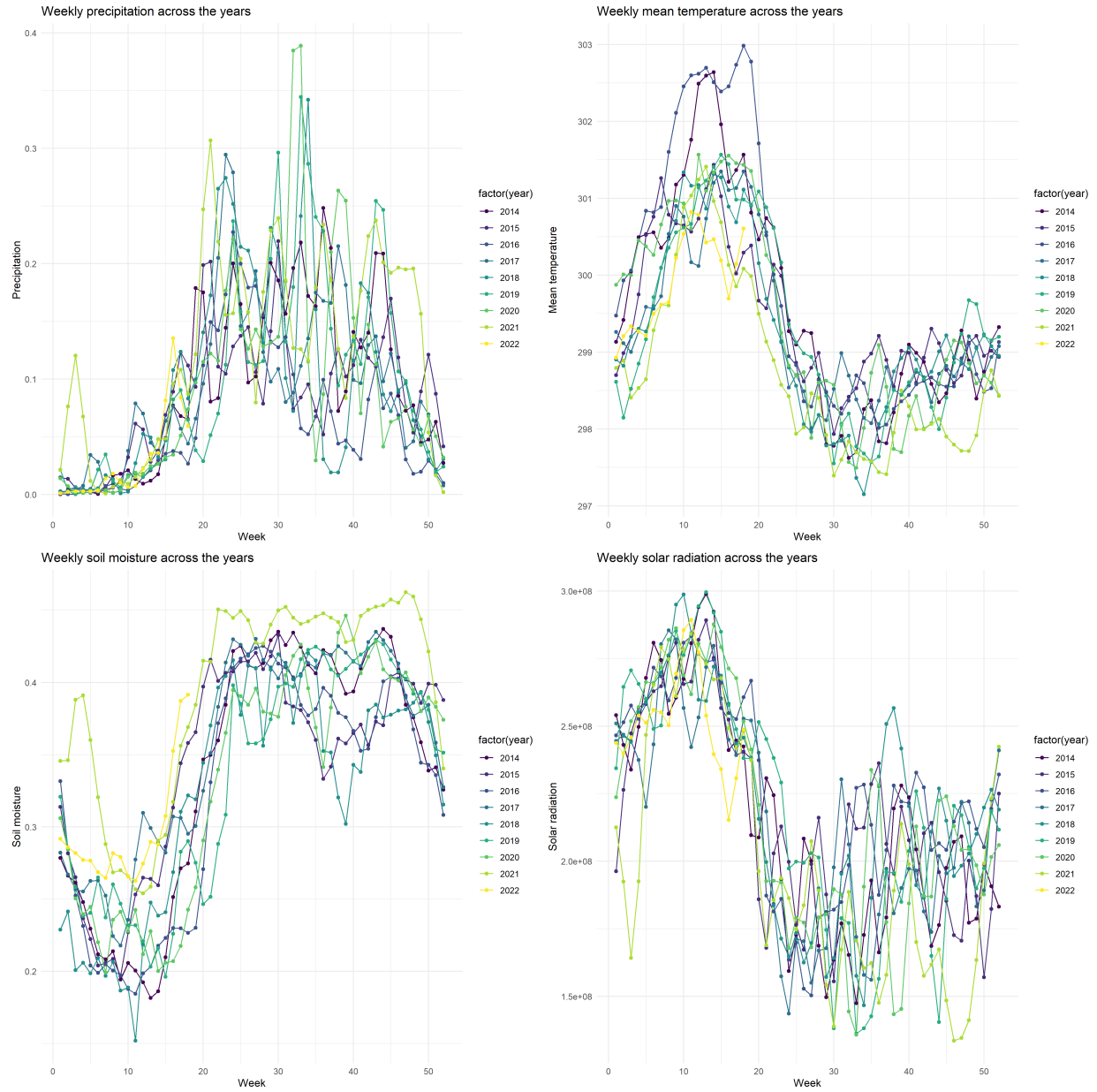


Figure S6. Mean weekly climate observations across locations where trees have been observed in Kerala through SeasonWatch

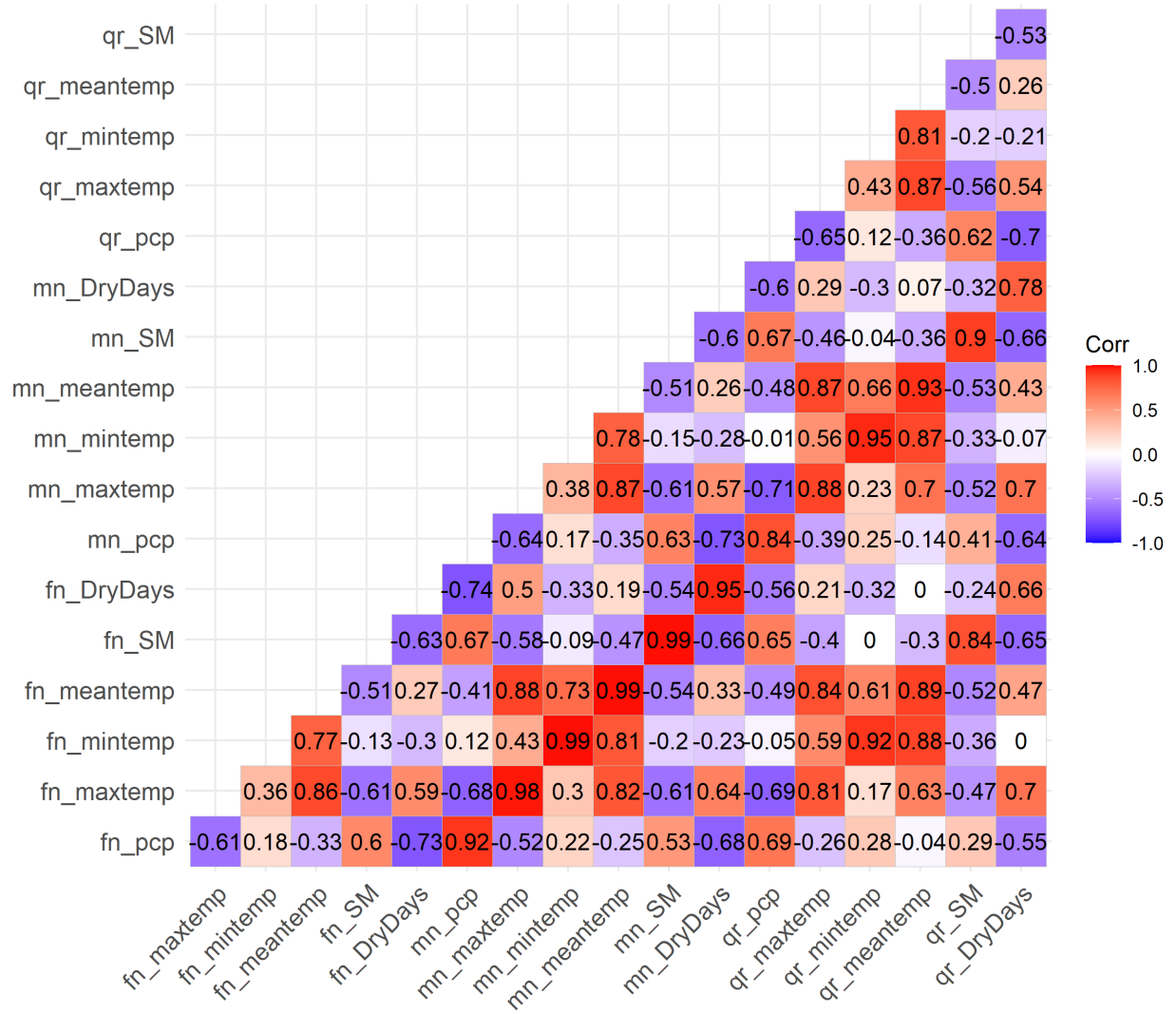


Figure S7. Correlations between environmental variables at different timescales from 2014-2022. Prefix fn_, mn_ and qr_ respectively denote fortnightly, monthly and quarterly variables. Variables are as follows: pcp = precipitation, maxtemp = maximum temperature, mintemp = minimum temperature, meantemp = mean temperature, SM = soil moisture, DryDays = number of consecutive dry days.

Section S4. Supplementary methods and results for ML modelling

For the entire set of observations, we used Random Forest models to classify the flowering occurrence observations based on environmental predictors. Random Forest classifier is a supervised learning algorithm that uses an ensemble of decision trees using majority voting to classify datasets into two categories; here the presence/absence of flowering (Breiman 2001). Random Forest classifiers have been used widely in ecological data analysis because of their flexibility and accuracy (Cutler et al. 2007). This approach allows for predicting the probability of observing flowering in a tree given recent environmental conditions.

For each of the three species, we ran separate models to classify flowering status according to the following model:

$$phenophase_{l,t} \sim precipitation_{l,t} + maximumtemperature_{l,t} + minimumtemperature_{l,t} + meantemperature_{l,t} + drydays_{l,t} + soilmoisture_{l,t} + solarradiation_{l,t}$$

Where precipitation, temperature, dry days, soil moisture, solar radiation are temporal predictors for the fortnight before the observation at location l and time t . We ran these models in R using the package *randomForest* (Liaw and Wiener 2002) for 500 decision trees. Tree-based methods are fairly insensitive to collinearity among predictors, allowing us to compare the full-range of temperature and precipitation variables simultaneously, as well as the full dataset, including repeat observations for the same tree.

We calculated variable importance for each model using the Mean Decrease in Accuracy (MDA) metric which removes a predictor from the model, re-trains the model to make predictions, and evaluates the decrease in accuracy.

Machine learning models with spatiotemporal predictors were able to classify flowering phenophases for the three species with substantial precision. The accuracy of a binary model by random chance would be 50%, leading to an F1 score of 0.5; models with higher F1 scores than randomly expected have predictive value. With 500 runs each, Random Forest models were able to predict the presence/absence of flowering in jackfruit, mango, and tamarind trees given the environmental conditions of the previous fortnight with F1 scores of 0.59, 0.62, and 0.26, respectively. These performance metrics describe the ability of the models to predict tree phenophases for an unobserved year based on environmental variables alone. We note that the distribution between flowering and no-flowering observations was about 2:1, potentially influencing the F1 score.

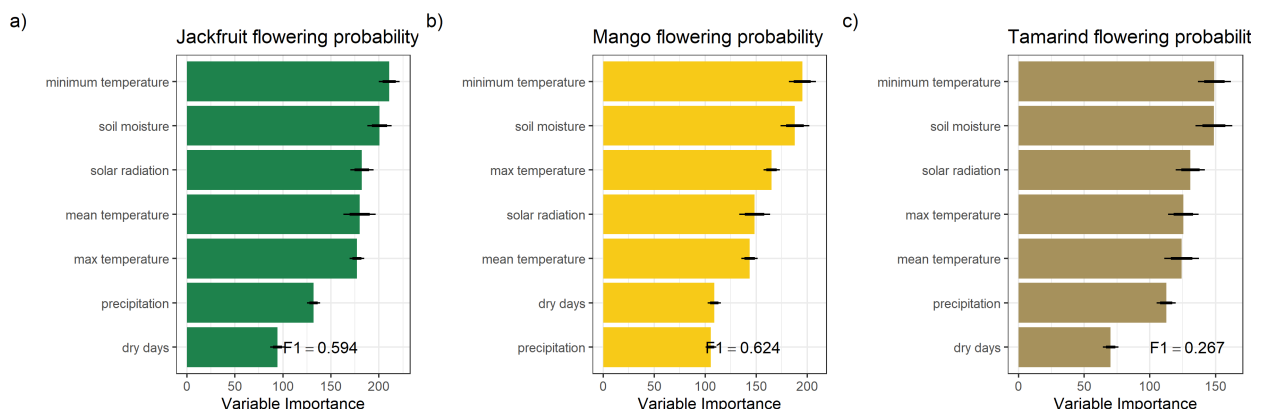


Figure S8. Variable importance for environmental variables at the fortnightly scale from machine learning models of flowering occurrence. Mean Decrease in Accuracy from Random Forest models of the probability of observing flowering in a) jackfruit, b) mango, c) tamarind. Variable importance has been calculated at each LOOCV run, Length of the bars represent mean MDA across the runs and variables are ordered in their decreasing order of mean importance. Thick black error bars represent 1 standard deviation from the mean and the thin bars represent 1.96 standard deviation from the mean across the runs.

Random forest models identified environmental variables strongly correlated with flowering for each of the species. The mean importance of each of the factors, calculated using MDA, was different across species.

For all three species, minimum temperature in the preceding fortnight was the most important predictor of flowering status. Among precipitation variables, soil moisture in the preceding fortnight had the strongest influence in classifying flowering status. Solar radiation also ranked among the top three variables for all three species. Across the 9 LOOCV runs, the relative importance of the variables had some variation, but total precipitation and the number of consecutive dry in the previous fortnight consistently were ranked lowest across runs in all species. Some of the variation in importance was also potentially influenced by the difference in number of observations across the years, with 2014 being the lowest sample.

Table S2. Variable importance across each of the LOOCV models for jackfruit

	mean temperature	minimum temperature	max temperature	solar radiation	precipitation	dry days	soil moisture
2014	194.2509	214.5788	179.2915	186.6246	135.1327	95.93955	194.1267
2015	179.5465	209.1443	188.5371	163.9275	126.3947	94.94915	198.4492
2016	190.2715	219.5325	175.8974	186.1020	132.2170	89.85891	202.5824
2017	165.7244	216.9539	172.4679	188.7910	135.5556	97.61727	215.0707
2018	163.4843	214.7208	175.2011	183.9784	127.4882	102.55036	195.5013
2019	184.4766	208.3534	176.8716	183.7102	139.4553	89.46958	207.0071
2020	184.7028	204.5251	174.5972	179.6201	133.1122	96.71695	207.2555
2021	173.7381	197.3297	173.6899	180.2438	130.7285	87.91403	189.8676
2022	182.7083	210.4814	177.0990	187.7085	127.7086	92.99067	195.6885

Table S3. Variable importance across each of the LOOCV models for mango

	mean temperature	minimum temperature	max temperature	solar radiation	precipitation	dry days	soil moisture
2014	134.4052	197.1949	164.8218	156.2773	104.72479	110.60015	202.8316
2015	144.9679	198.7177	163.3524	141.5846	109.64898	114.76737	180.4693
2016	144.6262	189.7379	155.3212	148.4978	107.66133	108.66425	200.1551
2017	144.4997	203.6345	167.7915	163.6967	110.57972	111.81441	187.1588
2018	138.2928	186.8528	163.5042	144.1055	103.84531	110.47167	178.7136
2019	140.8394	193.5883	163.4009	143.2801	99.64859	108.26231	186.1939
2020	144.5293	188.2500	169.6981	138.2137	106.36116	107.26544	191.2922
2021	148.4355	187.6162	163.8984	139.5365	104.15356	99.78337	178.4911
2022	150.8697	211.2650	173.2673	160.6331	103.66976	108.27752	185.9629

Table S4. Variable importance across each of the LOOCV models for tamarind

	mean temperature	minimum temperature	max temperature	solar radiation	precipitation	dry days	soil moisture
2014	125.3954	157.0129	123.5367	137.5659	112.1205	73.43858	161.9501
2015	112.7048	134.8631	119.5150	127.1199	108.4221	71.68865	145.6994
2016	133.7776	161.3472	123.6452	131.0377	109.8208	69.80641	152.8163
2017	129.6894	149.9381	122.4582	129.9581	112.6012	71.56924	154.0277
2018	112.3716	150.3709	129.8349	117.4087	105.5565	69.48091	149.1446
2019	117.0161	152.5773	114.9969	126.8927	117.6458	67.53726	144.7113
2020	124.7591	143.0515	131.3704	135.2399	119.6935	75.25158	131.8492
2021	131.1785	148.6381	124.0858	131.2129	110.6177	62.53390	155.7232
2022	131.0931	144.1736	140.0442	141.4173	116.0824	69.89865	143.3540

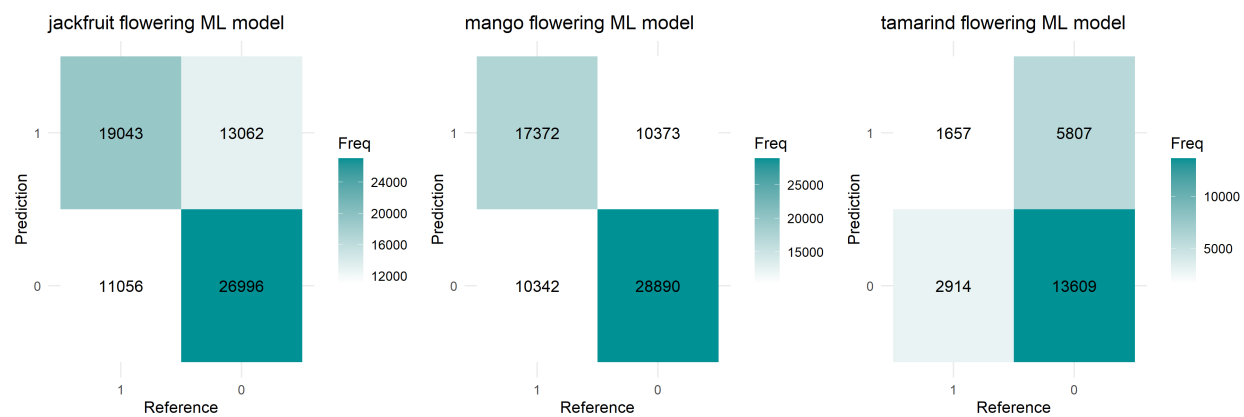


Figure S9. Confusion matrices with Random Forest models

Section S5. Supplementary results for GLMMs

Table S5. Model coefficients for flowering onset in jackfruit

# Fixed Effects					
Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	-3.13	0.11	(-3.35, -2.92)	-28.58	< .001
min temp	-0.65	0.05	(-0.75, -0.56)	-13.62	< .001
soil moisture	-0.26	0.08	(-0.41, -0.11)	-3.30	< .001
solar rad	0.07	0.05	(-0.03, 0.18)	1.37	0.169
elev cs	-0.17	0.06	(-0.29, -0.05)	-2.73	0.006
aspect cs	-0.08	0.04	(-0.16, 6.07e-03)	-1.82	0.069
lights cs	0.06	0.06	(-0.06, 0.18)	0.92	0.356
soil moisture $\times$ elev cs	-0.04	0.09	(-0.22, 0.13)	-0.49	0.622
solar rad $\times$ elev cs	-0.10	0.05	(-0.19, -3.39e-03)	-2.03	0.042
min temp $\times$ aspect cs	-0.03	0.03	(-0.09, 0.03)	-1.07	0.285
soil moisture $\times$ aspect cs	-0.03	0.04	(-0.10, 0.04)	-0.78	0.433
solar rad $\times$ lights cs	0.06	0.03	(-1.77e-03, 0.12)	1.90	0.057

Random Effects

Parameter	Coefficient
SD (Intercept: Grid_ID)	0.80

Table S6. Model coefficients for flowering onset in mango

# Fixed Effects					
Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	-3.08	0.10	(-3.28, -2.88)	-30.62	< .001
min temp	-0.75	0.05	(-0.85, -0.65)	-14.59	< .001
soil moisture	-0.20	0.08	(-0.35, -0.05)	-2.65	0.008
solar rad	0.19	0.06	(0.08, 0.30)	3.40	< .001
elev cs	-0.34	0.07	(-0.48, -0.19)	-4.60	< .001
aspect cs	-0.03	0.04	(-0.12, 0.06)	-0.71	0.478
lights cs	0.16	0.06	(0.05, 0.27)	2.87	0.004
soil moisture $\times$ elev cs	0.20	0.09	(0.02, 0.38)	2.14	0.032
solar rad $\times$ elev cs	0.05	0.06	(-0.06, 0.16)	0.97	0.334
min temp $\times$ aspect cs	0.02	0.03	(-0.05, 0.08)	0.51	0.609
soil moisture $\times$ aspect cs	-0.01	0.04	(-0.08, 0.06)	-0.31	0.754
solar rad $\times$ lights cs	-0.06	0.03	(-0.12, 2.38e-03)	-1.88	0.060

Random Effects

Parameter	Coefficient
SD (Intercept: Grid_ID)	0.78

Table S7. Model coefficients for flowering onset in tamarind

Fixed Effects

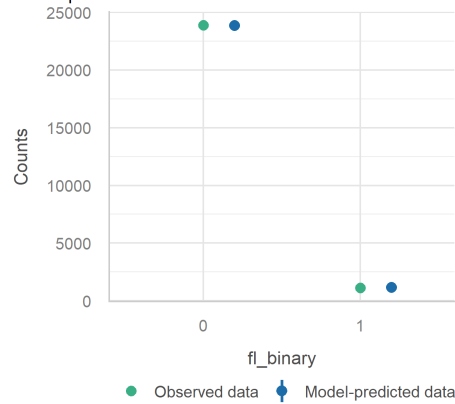
Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	-3.28	0.13	(-3.53, -3.02)	-25.21	< .001
min temp	0.23	0.08	(0.07, 0.38)	2.82	0.005
soil moisture	0.30	0.14	(0.03, 0.58)	2.14	0.032
solar rad	0.06	0.09	(-0.12, 0.24)	0.65	0.517
elev cs	-0.05	0.16	(-0.36, 0.26)	-0.32	0.752
aspect cs	-0.08	0.07	(-0.22, 0.07)	-1.02	0.306
lights cs	-8.52e-04	0.07	(-0.15, 0.14)	-0.01	0.991
soil moisture × elev cs	0.12	0.17	(-0.21, 0.45)	0.71	0.479
solar rad × elev cs	0.07	0.08	(-0.09, 0.23)	0.89	0.373
min temp × aspect cs	-0.03	0.06	(-0.14, 0.08)	-0.58	0.562
soil moisture × aspect cs	-0.04	0.07	(-0.17, 0.09)	-0.58	0.563
solar rad × lights cs	2.35e-04	0.05	(-0.09, 0.09)	4.96e-03	0.996

Random Effects

Parameter	Coefficient
SD (Intercept: Grid_ID)	0.63

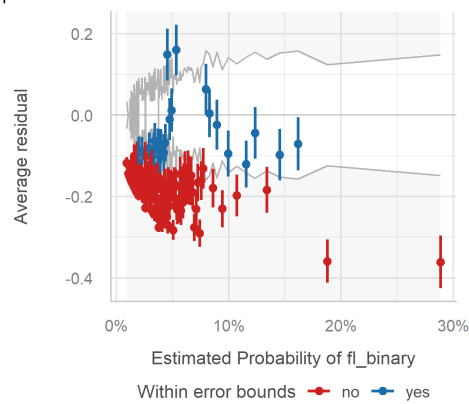
Posterior Predictive Check

Model-predicted intervals should include observed data points



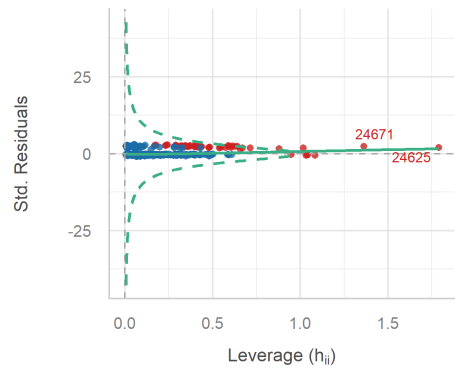
Binned Residuals

Points should be within error bounds



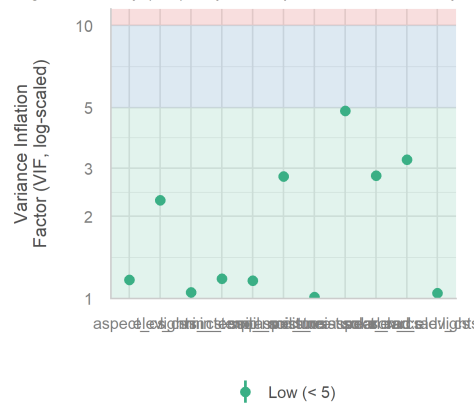
Influential Observations

Points should be inside the contour lines



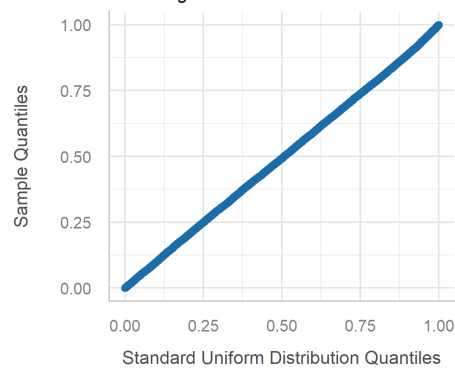
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



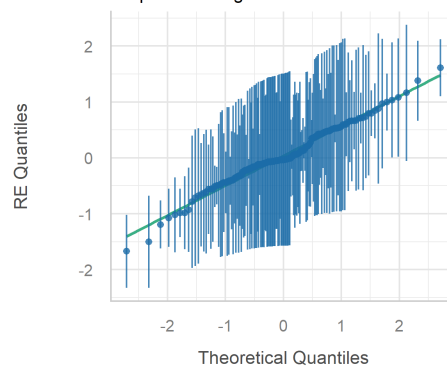
Distribution of Quantile Residuals

Dots should fall along the line



Normality of Random Effects (Grid_ID)

Dots should be plotted along the line

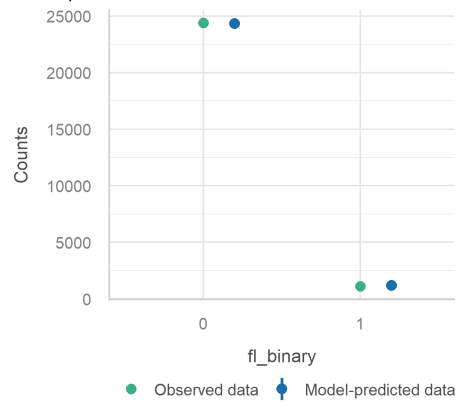


agnostics for onset of flowering in jackfruit

Figure S10. GLMM diag-

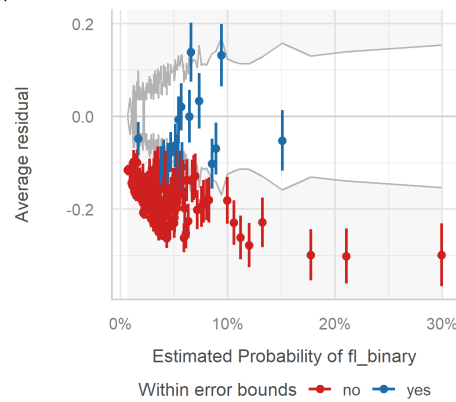
Posterior Predictive Check

Model-predicted intervals should include observed data points



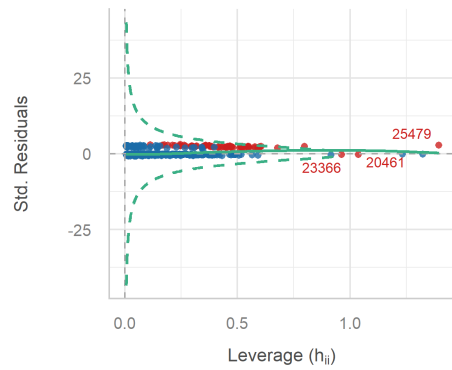
Binned Residuals

Points should be within error bounds



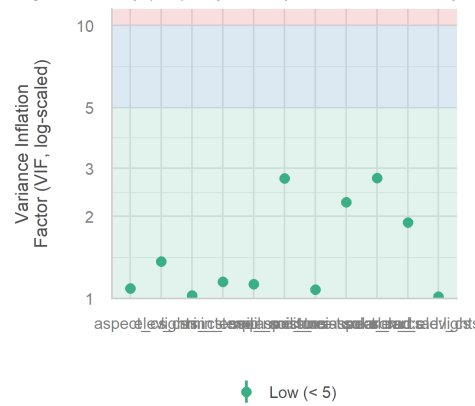
Influential Observations

Points should be inside the contour lines



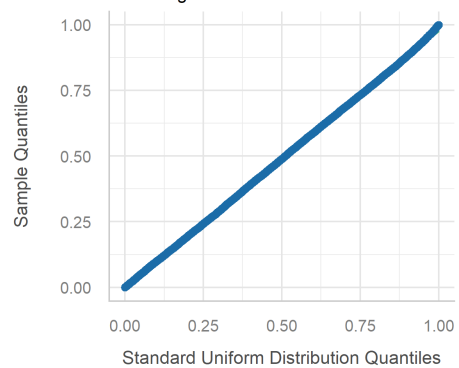
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



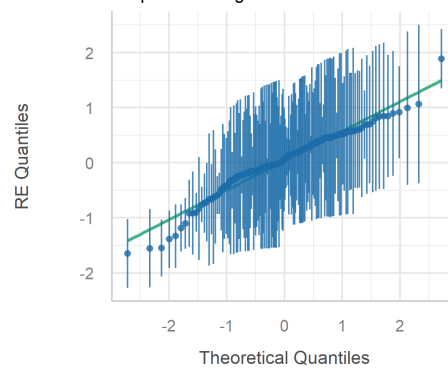
Distribution of Quantile Residuals

Dots should fall along the line



Normality of Random Effects (Grid_ID)

Dots should be plotted along the line

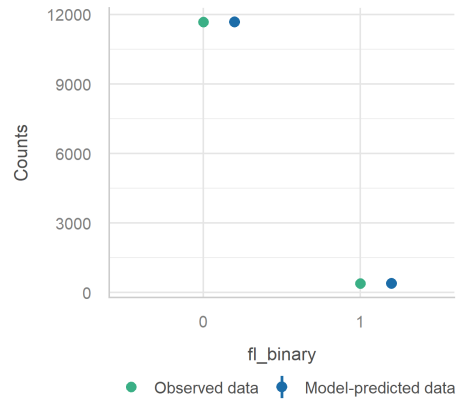


tics for onset of flowering in mango

Figure 11. GLMM diagnos-

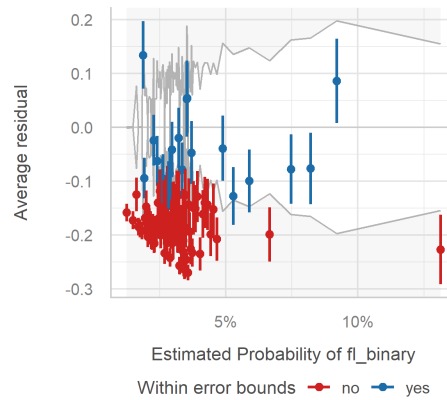
Posterior Predictive Check

Model-predicted intervals should include observed data points



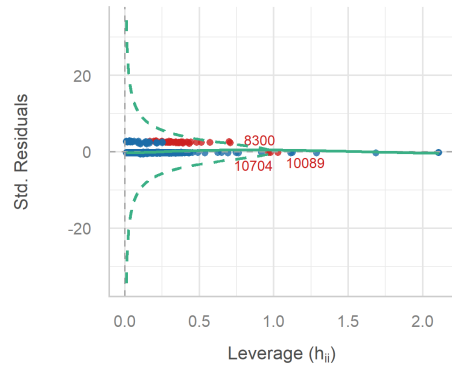
Binned Residuals

Points should be within error bounds



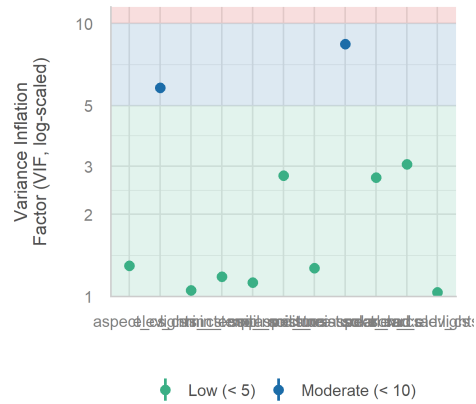
Influential Observations

Points should be inside the contour lines



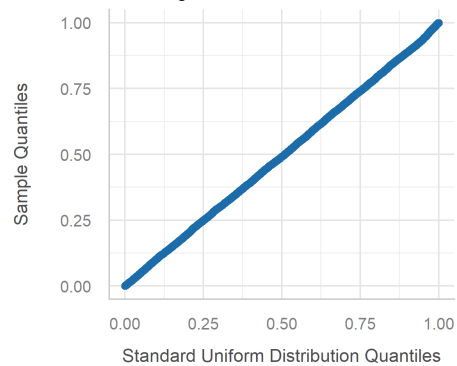
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



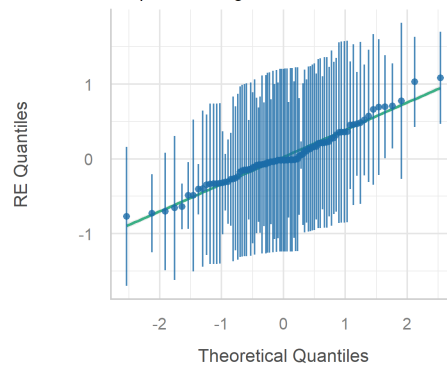
Distribution of Quantile Residuals

Dots should fall along the line



Normality of Random Effects (Grid_ID)

Dots should be plotted along the line



agnostics for onset of flowering in tamarind

Figure S12. GLMM diag-

Section S6. Leave-one-out cross validation for GLMMs

Table S8. Model coefficients for LOOCV models of flowering onset in jackfruit

var	2014	2015	2016	2017	2018	2019	2020	2021	2022
(Intercept)	-3.19 (0.11)	-3.16 (0.11)	-3.19 (0.12)	-3.14 (0.11)	-3.07 (0.11)	-3.19 (0.12)	-3.08 (0.11)	-3.08 (0.11)	-3.12 (0.11)
min_temp	-0.65 (0.05)	-0.64 (0.05)	-0.68 (0.05)	-0.66 (0.05)	-0.63 (0.05)	-0.72 (0.06)	-0.59 (0.05)	-0.61 (0.05)	-0.65 (0.05)
soil_moisture	-0.21 (0.08)	-0.26 (0.08)	-0.22 (0.08)	-0.25 (0.08)	-0.38 (0.09)	-0.29 (0.09)	-0.24 (0.08)	-0.19 (0.08)	-0.28 (0.08)
solar_rad	0.09 (0.06)	0.08 (0.06)	0.06 (0.06)	0.08 (0.06)	0 (0.06)	0.05 (0.06)	0.1 (0.06)	0.18 (0.06)	0.07 (0.05)
elev_cs	-0.15 (0.06)	-0.17 (0.07)	-0.12 (0.06)	-0.15 (0.06)	-0.27 (0.09)	-0.19 (0.07)	-0.15 (0.06)	-0.19 (0.07)	-0.17 (0.06)
aspect_cs	-0.06 (0.04)	-0.1 (0.05)	-0.08 (0.04)	-0.08 (0.04)	-0.05 (0.05)	-0.09 (0.05)	-0.05 (0.05)	-0.14 (0.05)	-0.07 (0.04)
lights_cs	0.06 (0.06)	0.03 (0.06)	0.03 (0.07)	0.05 (0.06)	0.06 (0.06)	0.08 (0.07)	0.12 (0.06)	0.05 (0.07)	0.05 (0.06)
soil_moisture:elev_cs	-0.08 (0.09)	-0.06 (0.1)	-0.13 (0.09)	-0.09 (0.1)	0.12 (0.11)	-0.07 (0.1)	0 (0.09)	-0.03 (0.1)	-0.04 (0.09)
solar_rad:elev_cs	-0.11 (0.05)	-0.08 (0.05)	-0.11 (0.05)	-0.13 (0.06)	-0.03 (0.05)	-0.12 (0.06)	-0.07 (0.05)	-0.13 (0.05)	-0.1 (0.05)
min_temp:aspect_cs	-0.03 (0.03)	-0.03 (0.03)	-0.03 (0.03)	-0.04 (0.03)	-0.03 (0.03)	-0.02 (0.03)	-0.03 (0.03)	-0.05 (0.03)	-0.04 (0.03)
soil_moisture:aspect_cs	-0.02 (0.04)	-0.01 (0.04)	-0.05 (0.04)	-0.02 (0.04)	0 (0.04)	-0.04 (0.04)	-0.02 (0.04)	-0.07 (0.04)	-0.04 (0.04)
solar_rad:lights_cs	0.06 (0.03)	0.09 (0.03)	0.04 (0.03)	0.08 (0.03)	0.04 (0.03)	0.07 (0.03)	0.07 (0.03)	0.02 (0.03)	0.06 (0.03)

Table S9. Model coefficients for LOOCV models of flowering onset in mango

var	2014	2015	2016	2017	2018	2019	2020	2021	2022
(Intercept)	-3.15 (0.1)	-3.08 (0.1)	-3.22 (0.11)	-3.03 (0.1)	-3.07 (0.1)	-3.12 (0.1)	-3.08 (0.11)	-3.11 (0.11)	-3.05 (0.1)
min_temp	-0.77 (0.05)	-0.73 (0.05)	-0.82 (0.06)	-0.71 (0.05)	-0.71 (0.05)	-0.79 (0.06)	-0.71 (0.05)	-0.71 (0.06)	-0.73 (0.05)
soil_moisture	-0.16 (0.08)	-0.12 (0.08)	-0.05 (0.08)	-0.16 (0.08)	-0.31 (0.08)	-0.25 (0.08)	-0.29 (0.08)	-0.22 (0.08)	-0.24 (0.08)
solar_rad	0.21 (0.06)	0.23 (0.06)	0.22 (0.06)	0.19 (0.06)	0.13 (0.06)	0.16 (0.06)	0.15 (0.06)	0.28 (0.07)	0.18 (0.06)
elev_cs	-0.34 (0.07)	-0.34 (0.07)	-0.44 (0.09)	-0.3 (0.07)	-0.34 (0.08)	-0.32 (0.07)	-0.29 (0.07)	-0.32 (0.1)	-0.35 (0.08)
aspect_cs	-0.03 (0.04)	-0.04 (0.05)	-0.04 (0.05)	-0.04 (0.04)	-0.06 (0.05)	0.01 (0.05)	-0.01 (0.05)	-0.05 (0.05)	-0.03 (0.04)
lights_cs	0.16 (0.06)	0.18 (0.06)	0.16 (0.06)	0.16 (0.06)	0.18 (0.06)	0.15 (0.06)	0.15 (0.06)	0.18 (0.06)	0.18 (0.06)

var	2014	2015	2016	2017	2018	2019	2020	2021	2022
soil_moisture:elev_cs	0.18 (0.09)	0.22 (0.1)	0.3 (0.11)	0.1 (0.1)	0.25 (0.1)	0.13 (0.1)	0.13 (0.09)	0.29 (0.1)	0.21 (0.09)
solar_rad:elev_cs	0.05 (0.06)	0.06 (0.06)	0.06 (0.06)	0.03 (0.06)	0.06 (0.06)	0.02 (0.06)	0.03 (0.06)	0.14 (0.07)	0.06 (0.06)
min_temp:aspect_cs	0.01 (0.03)	0.01 (0.04)	-0.01 (0.04)	-0.01 (0.04)	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.01 (0.04)	0.01 (0.03)
soil_moisture:aspect_cs	-0.01 (0.04)	0.01 (0.04)	-0.04 (0.04)	-0.02 (0.04)	0.01 (0.04)	-0.02 (0.04)	0.01 (0.04)	-0.05 (0.04)	-0.01 (0.04)
solar_rad:lights_cs	-0.06 (0.03)	-0.04 (0.03)	-0.08 (0.03)	-0.04 (0.03)	-0.06 (0.03)	-0.06 (0.03)	-0.05 (0.03)	-0.13 (0.04)	-0.04 (0.03)

Table S10. Model coefficients for LOOCV models of flowering onset in tamarind

var	2014	2015	2016	2017	2018	2019	2020	2021	2022
(Intercept)	-3.38 (0.14)	-3.37 (0.14)	-3.27 (0.14)	-3.29 (0.13)	-3.26 (0.14)	-3.31 (0.14)	-3.29 (0.14)	-3.15 (0.14)	-3.23 (0.13)
min_temp	0.18 (0.08)	0.19 (0.08)	0.24 (0.09)	0.2 (0.08)	0.24 (0.09)	0.43 (0.1)	0.25 (0.08)	0.14 (0.09)	0.2 (0.08)
soil_moisture	0.39 (0.15)	0.29 (0.15)	0.34 (0.15)	0.25 (0.14)	0.21 (0.15)	0.45 (0.16)	0.37 (0.15)	0.18 (0.15)	0.27 (0.15)
solar_rad	0.1 (0.09)	0.1 (0.1)	0.1 (0.09)	-0.02 (0.09)	0.08 (0.1)	0 (0.1)	0.15 (0.1)	-0.08 (0.1)	0.05 (0.09)
elev_cs	-0.05 (0.16)	-0.13 (0.18)	0.02 (0.15)	-0.13 (0.18)	-0.04 (0.17)	-0.01 (0.18)	-0.06 (0.17)	-0.03 (0.2)	-0.04 (0.16)
aspect_cs	-0.09 (0.08)	-0.03 (0.08)	-0.13 (0.08)	-0.05 (0.08)	-0.09 (0.08)	-0.1 (0.08)	-0.1 (0.08)	-0.05 (0.08)	-0.06 (0.08)
lights_cs	-0.02 (0.08)	-0.02 (0.08)	0.02 (0.08)	0 (0.07)	-0.03 (0.08)	0.01 (0.08)	-0.04 (0.08)	0.02 (0.08)	0.02 (0.08)
soil_moisture:elev_cs	0.1 (0.17)	0.21 (0.18)	0.05 (0.17)	0.18 (0.18)	0.09 (0.18)	0.14 (0.18)	0.15 (0.17)	0.02 (0.26)	0.1 (0.17)
solar_rad:elev_cs	0.06 (0.08)	0.1 (0.08)	0.05 (0.08)	0.09 (0.08)	0.07 (0.09)	0.08 (0.08)	0.07 (0.08)	0.03 (0.17)	0.07 (0.08)
min_temp:aspect_cs	-0.05 (0.06)	-0.02 (0.06)	-0.09 (0.06)	0 (0.06)	-0.02 (0.06)	0 (0.06)	-0.05 (0.06)	-0.05 (0.06)	-0.03 (0.06)
soil_moisture:aspect_cs	-0.03 (0.07)	-0.07 (0.07)	-0.06 (0.07)	-0.07 (0.07)	-0.01 (0.07)	-0.01 (0.07)	-0.03 (0.07)	-0.04 (0.07)	-0.04 (0.07)
solar_rad:lights_cs	-0.02 (0.05)	0 (0.05)	0 (0.05)	-0.01 (0.05)	0.02 (0.05)	0.05 (0.05)	-0.03 (0.05)	0.01 (0.06)	0.01 (0.05)

Table S11. Out-of-bag predictive power for LOOCV models of flowering onset in jackfruit

year	cor	mse
2014	0.2004693	0.0584499
2015	0.1217795	0.0432738
2016	0.1265467	0.0467056
2017	0.0935850	0.0384501
2018	0.1332462	0.0518880
2019	0.1155356	0.0457906
2020	0.0905950	0.0340696
2021	0.0539513	0.0320582
2022	0.1485688	0.0601856

Table S12. Out-of-bag predictive power for LOOCV models of flowering onset in mango

year	cor	mse
2014	0.1026427	0.0654172
2015	0.1699537	0.0388611
2016	0.0948727	0.0517563
2017	0.1589169	0.0341097
2018	0.1419168	0.0555414
2019	0.1498177	0.0356736
2020	0.1218867	0.0416410
2021	0.0684566	0.0334110
2022	0.2267429	0.0428476

Table S13. Out-of-bag predictive power for LOOCV models of flowering onset in tamarind

year	cor	mse
2014	0.0235131	0.0668460
2015	0.1197022	0.0668705
2016	0.0557611	0.0337775
2017	0.1026259	0.0303872
2018	0.0479079	0.0454108
2019	0.0025457	0.0307121
2020	-0.0085910	0.0246159
2021	0.0165670	0.0262515
2022	0.0621319	0.0262394

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